



Course Outline

1. Course Identity

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	CSC3100
Course title (English)	Data Structures
Course title (Chinese)	数据结构
Units	3
Language of Instruction	English
Description (English)	The concept of abstract data types and the advantages of data abstraction are introduced. Various commonly used data types including array, stack, queue, list, tree, and graph, and their implementations will be discussed. In addition, advanced data structures such as iterator and heap will be covered. Applications including basic sorting and searching methods are described to illustrate the use of data structures, together with the complexity analysis of the algorithms.
Description (Chinese)	本课程将介绍抽象数据类型的概念和数据抽象的优点。讨论各种常用的数据类型，包括数组、堆栈、队列、列表、树和图，以及它们的实现。此外，本课程还将介绍迭代器和堆等高级数据结构，并以排序和搜索为例简要介绍数据结构的应用和算法的复杂性分析。

2. Prerequisites / Co-requisites

A. Prerequisites

(1) CSC1001 Introduction to Computer Science: Programming Methodology
or (2) CSC1003 Introduction to Computer Science and Java Programming

B. Co-requisites

3. Learning Outcomes



Upon completing this course, students will be able to:

- ☐ Understand the important role of data structure in computer science (K)
- ☐ Grasp the concepts and implementations of various data structures and the associated operations (K & S)
- ☐ Understand the advantages and disadvantages of each data structure covered in the course (K)
- ☐ Apply several key and fundamental algorithms (K & S)
- ☐ Analyze a given algorithm in terms of correctness and complexity (K & S)
- ☐ Get prepared for further advanced study of data structures and algorithms, and more general topics in computer science (V)

4. Course syllabus

- 1.Introduction
 - 1.1Java Basics
 - 1.2Complexity Analysis
- 2.Arrays
 - 2.1Basic Concepts
 - 2.2Implementation
 - 2.3Examples
- 3.Lists
 - 3.1Basic Concepts
 - 3.2Implementation
 - 3.3Examples
- 4.Stacks and Queues
 - 4.1Basic Concepts
 - 4.2Implementation
 - 4.3Examples
- 5.Trees
 - 5.1Basic Concepts
 - 5.2Implementation
 - 5.3Examples
- 6.Graphs
 - 6.1Basic Concepts
 - 6.2Implementation
 - 6.3Examples
- 7.Others: Iterators, Heaps, Maps, Sets
 - 7.1Basic Concepts
 - 7.2Implementation
 - 7.3Examples



- 8.Hashing
- 8.1Basic Concepts
- 8.2Implementation
- 8.3Examples
- 9.Applications
- 9.1Sorting
- 9.2Searching

5. Assessment Scheme

Component/ method	% weight
Homework Assignments	39
Midterm Exam	20
Final Exam	40
Course Evaluation	1

6. Grade descriptor

General

Grade	Description
A	Outstanding performance on learning all the data structures, algorithms, and analysis methods covered in this course. Show exceptional ability to choose proper data structures for solving different tasks. Be able to analyze the correctness and complexity of an arbitrarily given algorithm and further design an algorithm with desired complexity. Show exceptional ability to use the algorithms learned to solve real world problems.
A-	Generally outstanding performance on learning most of the data structures, algorithms, and analysis methods covered in this course. Show good ability to choose proper data structures for solving different tasks. Be able to analyze the correctness and complexity of an arbitrarily given algorithm. Show good ability to use the algorithms learned to solve real world problems.
B	Good performance on learning most of the data structures, algorithms, and analysis methods covered in this course. Show good ability to choose proper data structures for solving different tasks. Be able to analyze the correctness and complexity of some given algorithms. Show good ability to use most of the algorithms learned to solve real world problems.
C	Good performance on learning many of the data structures, algorithms, and analysis methods covered in this course. Show the ability to distinguish the various data structures and may choose proper data structures for solving some tasks. Be able to analyze the correctness and complexity of some given algorithms. Show good ability to use some of the algorithms learned to solve real world problems.



D	Good performance on learning many of the simple data structures, algorithms, and analysis methods covered in this course. Show the ability to distinguish the commonly used elementary data structures and may choose proper data structures for solving some simple tasks. Be able to analyze the correctness and complexity of some simple algorithms. Show good ability to use some of the simple algorithms learned to solve real world problems.
F	Unsatisfactory performance on learning many of the data structures, algorithms, and analysis methods covered in this course. Fail to distinguish the commonly used elementary data structures. Not be able to analyze the correctness and complexity of some simple algorithms. Have poor ability to use the algorithms learned to solve real world problems.

7. Feedback for evaluation

- Informal feedback to the instructor and/or teaching assistant(s) through face-to-face talking, phone calls, emails, the course website, etc.;
- Advisor-advisee program and programme review

8. Reading

Required Reading

1.Introduction to Algorithms, 978-0262033848, T.H. Cormen, C.E. Leiserson, R.L. Rivest, C. Stein, 3RD EDITION, United States

9. Course components

Activity	Hours/week
Lecture	3
Reading & Assignments	6
Tutorial	1

10. Indicative teaching plan

Week	Content/ topic/ activity
001	Introduction, Java basics
002	Arrays, time complexity
003	Insertion sort, merge sort



004	List
005	Stack, queues
006	Sorting algorithms
007	Trees
008	Midterm test
009	Trees
010	Hashing
011	Graphs
012	Graphs
013	Graphs, other data structures
014	Course review
	Course review
	Course review
	Course review

11. Any other information

None.

12. Version date

Version number	7
As of (date)	2024/05/29